AI Speech Diarization

1290_CSEN: Kenneth Kang
Advisor: Sean Choi

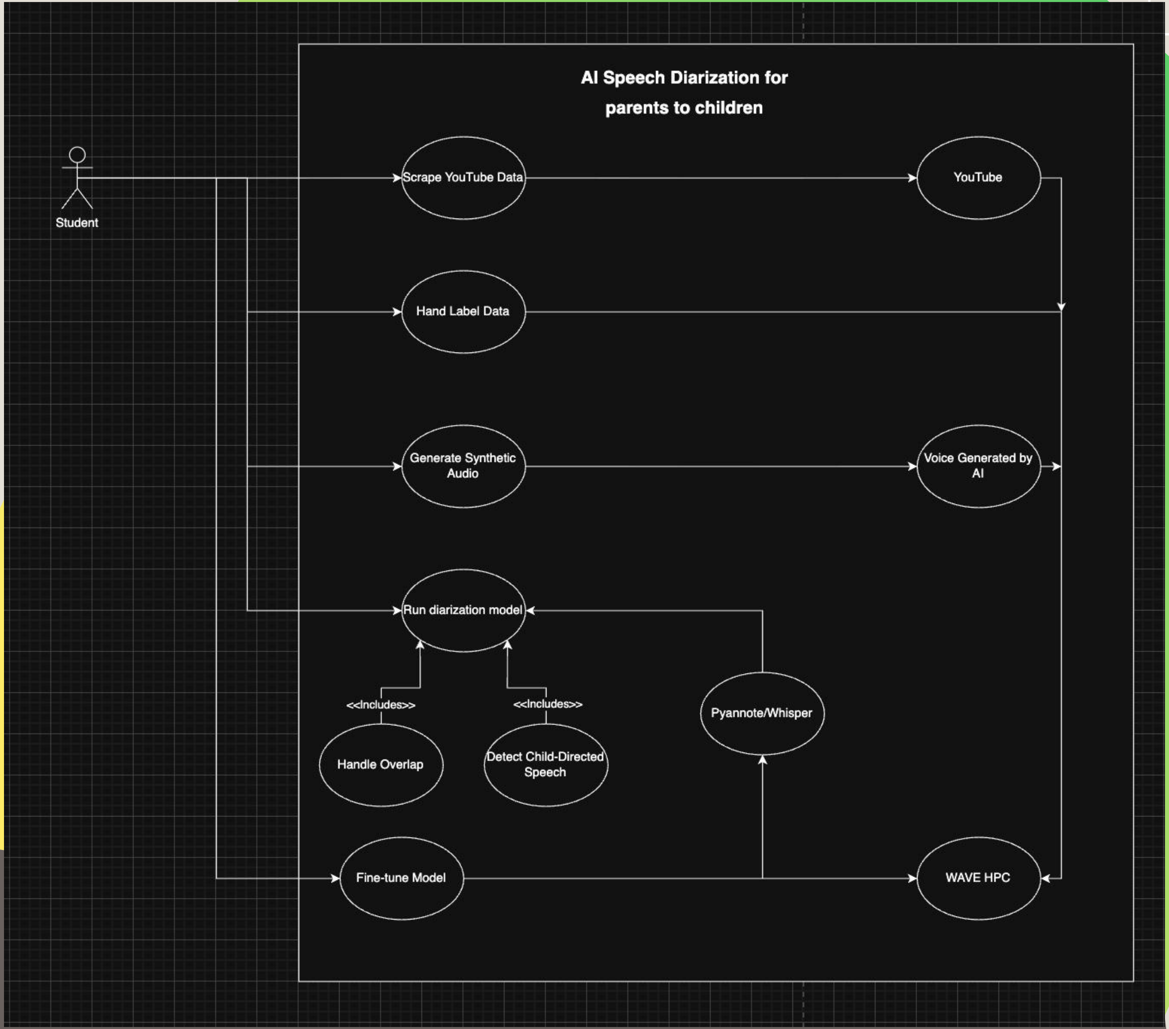
Background

AI Speech Diarization for parents to children

Speech diarization programs in many situations tend to misinterpret speech patterns of parents talking to children. In many cases, inaccurately labeling due to code switching.

Programs such as these have found success in interpreting adult to adult speech but using it for parent to child speech is the next step in improving speech diarization. In many cases improving diarization will also improve the quality of audio indexing, education, law enforcement, and many other fields.

Use Cases



AI Speech Diarization for
parents to children
Student
Scrape YouTube Data
YouTube
Hand Label Data
Generate Synthetic Audio
Voice Generated by AI
Run diarization model
<<Includes>>
<<Includes>>
Handle Overlap
Detect Child-Directed Speech
Pyannote/Whisper
Fine-tune Model
WAVE HPC

Goals

Test

Test what yields the best results and how AI speech Diarization can be improved for future use.

Target

Research and develop of AI speech Diarization that can take parent to child conversations

Stretch

Get diarization to parse for multiple people or multilingual conversations

Project Requirements

Functional Requirements

- **Automated Data Collection:** The system must scrape YouTube for parent-child interaction videos to create a training dataset.
- **Specialized Diarization:** The system must accurately distinguish and label "Child-Directed Speech," "Child Speech," and "Overlapping Speech".
- **Structured Output:** The system must generate a timestamped log identifying who is speaking (e.g., "Parent," "Child") and when.

Nonfunctional Requirements

- **Improved Accuracy:** The model must demonstrate better performance than the standard Pyannote 3.1
- **HPC Compatibility:** The training and fine-tuning processes must be configured to run on the SCU WAVE HPC cluster.
- **Cost Efficiency:** The project must rely on public/scraped data to avoid the high cost of traditional in-person data collection.

Constraints

Computational Resources:

Using HPC WAVE offered by SCU.

Project Schedule:

The development starts in the middle of Winter quarter due to group issues during Fall quarter.

Data Sourcing:

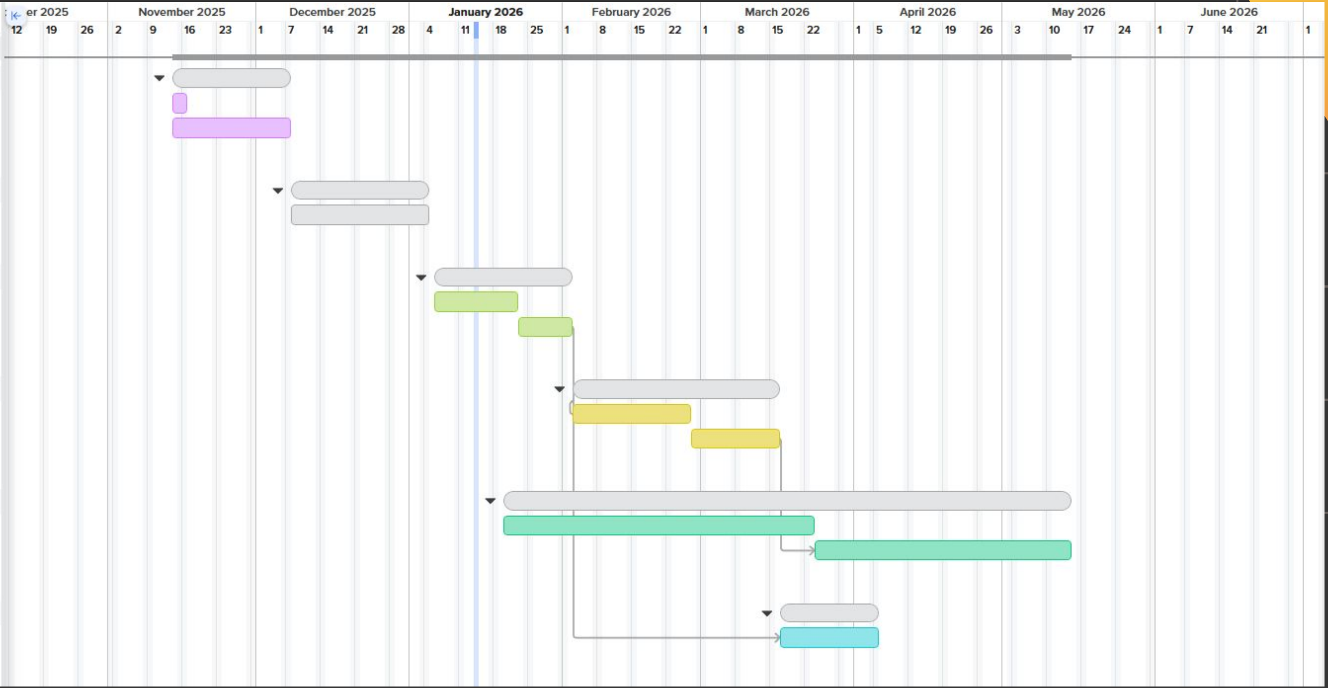
To remain cost-effective, the project is constrained to using publicly available data.

Standards

1	**ISO/IEC 27001: Information Security Management**	This is to ensure secure data when storing. The audio and files should be securely on WAVE HPC cluster and data will be limited to certain users.
2	**ISO/IEC 42001: Artificial Intelligence Management**	Standard for AI responsibility which legitimizes our work as a “Research Project”.
3	**NIST AI Risk Management Framework (AI RMF)**	Standard for US based research to make sure AI is safe and reliable. Helps address “Validity” and “ Reliability” making sure that metrics are trustworthy too.

Timeline

	Start date	End date
1290_CSEN_MLSpeechDia...	**11/14/25**	**5/14/26**
Initial Phase	**11/14/25**	**12/7/25**
Contact Wave HPC	11/14/25	11/16/25
Initial Research	11/14/25	12/7/25
Project Gap	**12/8/25**	**1/4/26**
No Activity	12/8/25	1/4/26
Baselines & Data	**1/6/26**	**2/2/26**
Test Existing products (Pyan...	1/6/26	1/22/26
Start Scraping YouTube/Lear...	1/23/26	2/2/26
Training Phase	**2/3/26**	**3/16/26**
Fine-Tuning	2/3/26	2/26/26
Revisions	2/27/26	3/16/26
Writing Thesis	**1/20/26**	**5/14/26**
Drafting	1/20/26	3/23/26
Finalizing	3/24/26	5/14/26
Final Testing	**3/17/26**	**4/5/26**
Validation	3/17/26	4/5/26



Test Plan

1. **Baseline Testing:** Run Pyannote and Whisper on videos of parent-children conversations talking over each. (Diarization Error Rate).

2. **Data Validation:** Check that the YouTube scraper successfully downloads clear audio and that your manual labels match the actual sound.

3. **Final Comparison:** Run your fine-tuned/modified model on the same videos used in baseline testing. Success is achieved if your model is better than the baseline.

Risks

WAVE Access:

If cluster access is delayed, you cannot fine-tune the model, making the 6-week training phase harder.

Model Limitations:

There is a risk that "fine-tuning" isn't enough to fix the overlap problem, as it is a fundamental weakness of current architecture.

Schedule Compression:

The 38-day gap (Dec-Jan) leaves almost no room for error; any delay now will miss the final deadline.

Technical Challenges

Overlapping Speech:

Differentiating two people talking at once is historically the hardest problem in diarization ("The Cocktail Party Problem")

Child Voice:

Standard models often confuse high-pitched "baby talk" or babbling with noise or silence, leading to missed data.

Data Noise:

YouTube clips are "wild data" which is much harder to process than clean lab recordings.

Ethical Concerns

1	**Privacy of Minors (Data Security):**	There is going to be use of YouTube scraping with audio of children which is a privacy concern. To minimize this we will anonymize the data used and have it under research only use cases.
2	**Consent & Copyright (Scraping):**	Although videos uploaded on YouTube are public there is concern of using that data for an experiment.
3	**Algorithmic Bias (AI Ethics)**	Having English only speaking parents may limit AI models ability also creating bias. To mitigate this we want to improve the AI first with parent to child interaction then introduce multilingual speech.

Thank you